

I DEWA MADE DHARMA PUTRA SANTIKA

IoT Software Engineer • Information Technology Student

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[LinkedIn](#) • [GitHub](#) • [Portfolio](#)

PROFILE

Information Technology student at ITB STIKOM Bali, focused on artificial intelligence, IoT, and robotics, with a growing interest in AI-driven software development. Background spans web and mobile products (React Native, Next.js), IoT integration (MQTT, Node-RED, Raspberry Pi/ESP32), applied research, and leading a student research group. Comfortable moving between technical and people-facing work, and generally aims for interfaces that stay clear and human-centered even as the systems behind them get more complex.

EXPERIENCE

IoT Development Intern – PT Bima Sakti Sanjaya, Denpasar, Bali Mar 2026 – Jul 2026

- Built a real-time web dashboard in Next.js for AgriSense, a smart farming IoT platform still in development, and supported the React Native mobile side of the project.
- Set up MQTT subscriptions and checked sensor telemetry end-to-end with Node-RED, connecting field sensors to a Raspberry Pi edge device.
- Designed the UI and data views for water (EC/TDS), air (temperature/humidity), and soil (moisture, pH, NPK) monitoring so the readings would be easier to act on.
- Joined senior engineers on site visits to inspect and maintain RTU Smart Panel installations – troubleshooting, checking wiring, verifying connectivity.
- Organized the collected sensor data with future AI use cases in mind, such as anomaly detection and predictive irrigation.

General Chair – RADE – Robotics and Embedded Systems Research Group, ITB STIKOM Bali Jan 2025 – Feb 2026

- Led cross-functional teams and kept project goals aligned with the group's broader objectives, which improved delivery and stakeholder satisfaction.
- Introduced stricter quality-assurance steps to keep research output consistent and reliable.
- Coordinated research activities, training sessions, seminars, and competitions across IoT, robotics, and AI – work that helped RADE earn recognition at the national level.

Research Assistant – Faculty Research – Hybrid K-NN for Dangdut Music Sub-Genre Classification 2024

- Contributed to a lecturer-led study applying a Hybrid K-Nearest Neighbour algorithm to classify dangdut music sub-genres.
- Helped with audio data collection, labeling, feature extraction (MFCC), and model evaluation.

FEATURED PROJECTS

SIGMA – Smart Intelligent Guidance for Mobility Automation – GEMASTIK 2025 Competition Project

- An IoT/AI traffic-signal system that uses YOLOv8 to detect vehicles and adjust green-light timing automatically, aimed at easing urban congestion; runs on edge computing (Raspberry Pi) with an analytics dashboard.

AgriSense Mobile – React Native Mobile App

- Real-time farm and greenhouse monitoring app with live sensor readings, remote actuator control, weather insights, AI-based recommendations, and historical data views in a clean, modern interface.

COASTHER – Smart Boarding House IoT System

- IoT system for smart boarding houses that tracks electricity use (ESP32, PZEM-004T) and water use (YF-S201) in real time over MQTT/HiveMQ.
- Built the monitoring dashboard in Next.js and supported a companion React Native app.

EDUCATION

Bachelor of Information Technology – ITB STIKOM Bali

Oct 2023 – May 2027 (expected)

Focused on artificial intelligence, IoT, and robotics, with active involvement in campus research and student organizations.

Student Exchange Program – Valaya Alongkorn Rajabhat University (VRU), Thailand, 2025: Cloud Computing, Project Management, IT Research Methodology.

SKILLS

Programming Languages: JavaScript/TypeScript, Python

Frontend & Mobile: React Native, Next.js, Tailwind CSS

AI & Machine Learning: TensorFlow, OpenCV, YOLOv8, Feature Engineering

IoT & Embedded Systems: Arduino/ESP32, Raspberry Pi, RTU, Node-RED, MQTT

Data & Infrastructure: MySQL, Firebase, GitHub

Soft Skills: Leadership & Team Management, Public Speaking, Strategic Planning, Problem Solving, Collaboration, Cross-Cultural Communication

CERTIFICATIONS

Android Development Associate – Logical Operations

Feb 2026

Credential ID 181045116 • TypeScript, CSS, and related skills.

Python Programming Associate – Logical Operations

Feb 2025

Python (Programming Language).

LANGUAGES

Indonesian: Native

English: B2 – Upper Intermediate

OTHER ACTIVITIES & ACHIEVEMENTS

- First place, poetry reading at PEKSIMIDA Bali, and Bali Province delegate at PEKSIMINAS 2024, representing ITB STIKOM Bali at the national level (Jakarta State University).
- Vice Chair, Teater Biner student theater group – wrote and performed poetry and theater pieces on social issues, 2023–2025.